Lab 7: Creating Reports and Dashboards in Splunk

Lab Overview

In this lab, you will build reports and dashboards for various stakeholders at Buttercup Games. You'll learn how to save searches as reports, create visualizations, build interactive dashboards, and manage permissions to share your work with others.

Learning Objectives

By the end of this lab, you will be able to:

- Create and save reports from search results
- Configure report permissions and sharing settings
- Build interactive dashboards with multiple panels
- Create different visualization types (charts, single values, tables)
- Arrange and customize dashboard layouts
- Use the visualization editor to select appropriate chart types
- Add panels to existing dashboards
- Navigate between search, visualization, and dashboard views

Prerequisites

- Completed Labs 1-5
- Understanding of Splunk search syntax and transforming commands
- Familiarity with stats functions (count, sum, avg, etc.)
- Basic knowledge of field extraction

Lab Environment Setup

Step 1: Verify Sample Data

Ensure you have the required sample data loaded:

```
index=main earliest=-30d latest=now
| stats count by sourcetype
```

Required sourcetypes:

- `access_combined_wcookie` - Web application logs

- `db_audit` - Database audit logs
- `linux_secure` - Linux security logs

Step 2: Search Settings

For all exercises:

1. Navigate to **Search & Reporting** app
2. Set time range to **Last 30 days**
3. Use the `main` index

> **Data window**: The course dataset covers roughly **26 days** ending at the moment `labs/data/generate_course_data.py` was last run — deliberately sized to sit inside Splunk's **Last 30 days** preset. Set the time picker to **Last 30 days** for every exercise unless a step says otherwise.
>
> Regenerate the data within **4 days** of delivering the class; past that the preset starts clipping the oldest events and the answer-key counts drift.

Key Concepts

Reports vs Dashboards

Reports

- Saved searches that can be run on demand
- Can include visualizations
- Shareable with specific permissions
- Can be scheduled for regular execution

Dashboards

- Collections of visualizations and reports
- Interactive panels with drill-down capabilities
- Real-time or scheduled data updates
- Customizable layouts and styling

Visualization Types

Type	Best For	Example Use Case
Column Chart	Comparing categories	Product sales by ID
Single Value	KPIs and metrics	Total sales count
Line Chart	Trends over time	Daily transaction volume
Pie Chart	Part-to-whole relationships	Traffic by browser type
Table	Detailed data display	IP addresses with attempt counts

Exercise 1: Security Report - Forbidden Access Attempts

Scenario: The security team needs a report of IP addresses attempting to access forbidden pages.

Task 1.1: Search for Forbidden Access

Find all web application events with forbidden status (403):

```
index=main sourcetype=access_combined_wcookie status=403
```

Expected Results: Events showing forbidden access attempts

Task 1.2: Count Attempts by IP

Add statistics to count attempts by client IP:

```
index=main sourcetype=access_combined_wcookie status=403
| stats count as attempts by clientip
```

Task 1.3: Sort Results

Display IPs with highest attempts first:

```
index=main sourcetype=access_combined_wcookie status=403
| stats count as attempts by clientip
| sort -attempts
```

✓ **Checkpoint**: Record the IP with most attempts: _____ (Total attempts: _____)

Task 1.4: Save as Report

1. Click **Save As** → **Report**

2. Enter report details:
    - **Title**: `403_by_clientip`
    - **Description**: `Security report showing IPs with forbidden access attempts`
    - **Content**: Statistics Table
    - **Time Range Picker**: Yes
3. Click **Save**

Task 1.5: Set Permissions

1. Click **Permissions** link
2. Configure settings:
    - **Display For**: App
    - **Run As**: Owner
    - **Read**: Everyone
    - **Write**: Power, Admin
3. Click **Save**

Task 1.6: Verify Report

1. Navigate to **Reports** menu
2. Find `403_by_clientip` in the list
3. Click to run the report

✓ **Checkpoint**: Report successfully created and accessible

Exercise 2: Sales Dashboard - Product Performance

Scenario: The CFO needs a dashboard showing product sales performance in one centralized location.

Task 2.1: Create Product Sales Visualization

Search for successful purchases:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
```

Task 2.2: Add Statistics

Count sales by product:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
| stats count by productId
```

Task 2.3: Create Column Chart

1. Click **Visualization** tab
2. Select **Column Chart** from Splunk Visualizations
3. Review the chart display

Task 2.4: Save to New Dashboard

1. Click **Save As** → **Dashboard Panel**
2. Configure settings:
   - **Dashboard**: New
   - **Dashboard Title**: `Sales Dashboard`
   - **Panel Title**: `Product Sales`
   - **Panel Description**: `Sales count by product ID`
3. Click **Save**
4. Click **View Dashboard**

✓ **Checkpoint**: Dashboard created with Product Sales panel

Task 2.5: Add Total Sales Panel

1. Click **Open in Search** icon at bottom of Product Sales panel
2. Modify search to remove the `by` clause:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
| stats count
```

3. Click **Visualization** tab
4. Select **Single Value** visualization

5. Click **Save As** → **Dashboard Panel**
6. Configure:
   - **Dashboard**: Existing
   - **Dashboard Title**: Sales Dashboard
   - **Panel Title**: `Total Units Sold`
7. Click **Save** → **View Dashboard**

Task 2.6: Arrange Dashboard Layout

1. Click **Edit** button on dashboard
2. Click and hold the bar at top of **Total Units Sold** panel
3. Drag panel to top position
4. Click **Save**

✓ **Checkpoint**: Dashboard displays Total Units Sold prominently at top

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Exercise 3: Enhanced Dashboard Panels

Scenario: Add more analytical panels to provide comprehensive sales insights.

Task 3.1: Top 5 Products Panel

Create a focused view of best sellers:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
| top limit=5 productId showperc=false
```

1. Select **Pie Chart** visualization
2. Save to existing Sales Dashboard
3. Panel Title: `Top 5 Products`

Task 3.2: Daily Sales Trend

Show sales over time:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
| timechart span=1d count as Sales
```

1. Select **Line Chart** visualization
2. Save to existing Sales Dashboard
3. Panel Title: `Daily Sales Trend`

Task 3.3: Customer Activity

Track unique customers:

```
index=main sourcetype=access_combined_wcookie
  file=success.do
  status=200
| stats dc(clientip) as "Unique Customers",
        count as "Total Purchases"
| eval "Avg Purchases per Customer"=round('Total Purchases'/'Unique Customers',2)
```

1. Select **Single Value** visualization with Trellis
2. Save to existing Sales Dashboard
3. Panel Title: `Customer Metrics`

Challenge Exercises

Challenge 1: Security Dashboard

Create a comprehensive security dashboard with:

1. **Failed Login Attempts Panel**:

```
index=main sourcetype=linux_secure "Failed password"
| rex "for (?:invalid user )?(?<user>\S+) from (?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count as Attempts by src_ip, user
| sort -Attempts
```

2. **404 Error Tracking**:

```
index=main sourcetype=access_combined_wcookie status=404
| timechart span=1h count
```

3. **Suspicious Activity Score**:

```
index=main sourcetype=access_combined_wcookie
| eval suspicious=case(
    status=403, 3,
    status=404, 1,
    file="oldlink", 2,
    1=1, 0)
| stats sum(suspicious) as "Risk Score" by clientip
| where 'Risk Score' > 10
```

Challenge 2: Performance Dashboard

Build a database performance dashboard:

1. **Slow Query Monitor**:

```
index=main sourcetype=db_audit Duration>1000
| stats count as "Slow Queries",
        avg(Duration) as "Avg Duration" by Command
| sort -"Avg Duration"
```

2. **Query Type Distribution**:

```
index=main sourcetype=db_audit
| stats count by Type
| eventstats sum(count) as total
| eval percent=round(count*100/total,2)
| fields - total
```

Challenge 3: Executive Dashboard

Create a C-suite dashboard with:

- Revenue metrics (simulated using bytes field)
- Customer engagement scores
- Operational KPIs
- Trend indicators with sparklines

Dashboard Best Practices

Layout Guidelines

1. **Priority Placement**: Most important metrics at top
2. **Logical Grouping**: Related panels together
3. **Visual Hierarchy**: Single values for KPIs, charts for trends
4. **Consistent Styling**: Uniform colors and formatting

Performance Optimization

1. **Base Searches**: Use post-process searches for related panels
2. **Time Range**: Set appropriate refresh intervals
3. **Data Models**: Consider using for frequently accessed data
4. **Scheduled Searches**: Pre-compute expensive searches

Sharing and Permissions

Permission Level	Access Rights
Private	Only creator can view
App	All app users can view
Global	Available across all apps

Troubleshooting Guide

Common Issues and Solutions

Issue	Solution
Report not appearing in list	Check permissions and app context
Visualization not displaying	Verify data format matches chart requirements
Dashboard panels not loading	Check search syntax and time range
Cannot edit dashboard	Verify write permissions
Single value shows N/A	Ensure search returns single numeric value

Dashboard Tips

- **Panel Linking**: Use drilldown to connect related dashboards
- **Input Controls**: Add time pickers and dropdowns for interactivity
- **Auto-Refresh**: Set appropriate refresh intervals for real-time data
- **Export Options**: Enable PDF/PNG export for sharing

Cleanup

This lab creates persistent objects. To clean up:

1. **Delete Reports**:
   - Navigate to Settings → Searches, reports, and alerts
   - Find and delete `403_by_clientip`
2. **Delete Dashboard**:
   - Navigate to Dashboards
   - Find and delete `Sales Dashboard`

Key Takeaways

✅ **You've learned to:**

- Create and manage reports with appropriate permissions
- Build multi-panel dashboards
- Select appropriate visualizations for different data types
- Arrange and customize dashboard layouts
- Share reports and dashboards with teams

Next Steps

- Proceed to Lab 8: Pivot Tables and Datasets
- Explore dashboard drilldown and form inputs
- Learn about scheduled reports and alerts

Quick Reference

Report Creation Workflow

1. Run search → Save As → Report
2. Set title and description
3. Configure permissions
4. Schedule if needed

Dashboard Panel Types

Panel Type	SPL Requirement	Best Practice
Single Value	Returns one numeric result	Use for KPIs
Chart	Returns series data	Include time or category field
Table	Returns tabular data	Limit rows for readability
Map	Returns geographic data	Include lat/lon or location fields

Visualization Selection Guide

```
Data Type → Recommended Visualization
├── Single Metric → Single Value
├── Categories → Column/Bar Chart
├── Time Series → Line/Area Chart
├── Part-to-Whole → Pie/Donut Chart
├── Correlation → Scatter Plot
└── Geographic → Cluster Map
```

Answer Key

🔑 **Instructor answer key**

Exercise 1

- IP with most forbidden (403) attempts: 92.46.53.223 (350 attempts)
- Total forbidden events: 1,936

Exercise 2

- Total units sold: 3,054
- Top selling product: WSC-MG-G10 (SIM Cubicle Tee)

Additional Metrics

- Cart additions: 11,405
- Purchases: 3,214
- Conversion rate: ~28.2% (3,214 purchases / 11,405 cart additions)
- Unique sessions (distinct JSESSIONID): 18,603